VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

CSV Data Processing System

A9513- Python Programming Laboratory

Submitted by

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VARDHAMAN
COLLEGE OF ENGINEERING

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Vibrant Vardhaman
25 Years of

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the Course End Project titled **"CSV Data Processing System"** is carried out by **Sahasra Voruganti, Shirisha Ande** with Roll Numbers **25881A05B3, 25881A05B9** towards **A9513 - Python Programming Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during the Academic year 2026-27.

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Abstract

The CSV Data Processing System is a modular Python application designed to import, validate, clean, transform, analyze, and export CSV (Comma-Separated Values) files. Built around the pandas and NumPy libraries, with Matplotlib for visualization and an optional Tkinter graphical interface, the system automates the routine but error-prone task of preparing raw tabular data for downstream analysis. Users can load a CSV file, review an automated data-quality report covering missing values, duplicate rows, and column data types, and then apply a configurable cleaning pipeline that imputes missing numeric values, standardizes categorical text, and removes duplicate records.

The system is organized into independent modules — Data Ingestion, Data Validator, Data Cleaner, Data Transformer, Report Generator, and Data Exporter — coordinated by a CSVProcessingPipeline class that also records performance metrics such as processing time and row/column counts before and after cleaning. Transformation utilities allow filtering, sorting, and group-wise aggregation, while the reporting module produces descriptive statistics and chart-based summaries (bar charts of missing-value counts and department-wise averages) that make data-quality issues and post-cleaning outcomes immediately visible. The cleaned dataset is written back out as a new CSV file, ready for use in spreadsheets, databases, or machine-learning pipelines.

Tested on a representative employee dataset containing missing ages, missing salaries, and a duplicate record, the system successfully reduced missing values from six to zero and duplicate rows from one to zero in under a hundredth of a second, while preserving all valid records. The project demonstrates how a small, well-structured Python toolkit can replace manual spreadsheet clean-up with a reliable, repeatable, and auditable process, making it a practical educational example of applied data engineering.

Keywords: CSV Processing, Data Cleaning, Data Validation, Data Transformation, Python, Pandas, Data Visualization, Automation, ETL, File Handling.

1 Introduction

The Comma-Separated Values (CSV) format has become the de-facto standard for exchanging tabular data between spreadsheets, databases, web services, and analytical tools. Its plain-text, human-readable structure makes it simple to generate and parse, which is precisely why it is used so widely — from exporting sales records and sensor logs to sharing survey responses and machine-learning training sets. However, this simplicity comes at a cost: CSV files carry no built-in schema, type information, or validation rules, so real-world CSV data is frequently messy. Missing fields, inconsistent formatting, duplicate rows, mismatched data types, and stray whitespace are common, and each of these issues can silently corrupt any analysis performed on top of the data.

Data preprocessing is widely recognized as one of the most time-consuming stages of any data analytics or machine-learning workflow, often estimated to consume the majority of a data professional's effort before any modelling or reporting can begin. Manually cleaning CSV files in a spreadsheet application does not scale well to large datasets, is difficult to reproduce, and is prone to human error. What is needed is a lightweight, automatable tool that can ingest a raw CSV file, systematically detect quality issues, apply consistent cleaning rules, and produce both a corrected dataset and a clear summary of what was changed and why.

This project, the CSV Data Processing System, addresses that need. It is a Python-based application that combines the data-manipulation power of the pandas library with a modular software design, so that ingestion, validation, cleaning, transformation, reporting, and export are each handled by a dedicated, independently testable component. The system is intended as both a practical utility for everyday CSV clean-up tasks and an educational demonstration of how software engineering principles — modularity, separation of concerns, and automated reporting — can be applied to a common data-engineering problem.

The remainder of this report reviews existing approaches to CSV data cleaning, states the project's objectives, describes the system's architecture and implementation in detail, presents results from processing a representative sample dataset, and maps the project's outcomes to relevant Program Outcomes and Sustainable Development Goals.

2 Literature Review

A range of tools and approaches already exist for cleaning and processing tabular data, each with trade-offs. Spreadsheet applications such as Microsoft Excel and Google Sheets offer accessible manual clean-up through filters, formulas, and macros, but they do not scale well to large files, make it difficult to reproduce a cleaning process exactly, and offer limited support for automated validation or logging of what was changed.

Programming-based approaches using the Python pandas library are the most common choice among data professionals, since pandas provides powerful, vectorized operations for handling missing values, duplicates, and type conversions. However, ad-hoc pandas scripts are typically written as a single, unstructured sequence of commands specific to one dataset, without a reusable architecture, automated data-quality reporting, or a clear separation between validation, cleaning, and transformation logic — limitations this project sets out to address through modular design.

Dedicated ETL (Extract-Transform-Load) and data-wrangling platforms, such as OpenRefine, Talend, and KNIME, offer graphical, rule-based data cleaning with strong auditing and transformation capabilities. These tools are powerful for enterprise data pipelines, but their configuration overhead and learning curve make them less suited to lightweight, single-file CSV clean-up tasks or educational settings where the internal logic of the cleaning process should be transparent and easy to inspect.

Recent academic and industry discussion has also emphasized automated data-quality reporting — summarizing missing-value counts, duplicate rates, and type inconsistencies before and after cleaning — as an essential complement to the cleaning operations themselves, since it allows users to verify that a pipeline behaved as expected rather than trusting it blindly. This project incorporates that principle directly: every cleaning run produces a metrics report (rows loaded, missing values before and after, duplicates before and after, and processing time) alongside visual summaries, closing the gap between simple pandas scripts and heavier commercial ETL suites.

By combining a modular, object-oriented pipeline with automated validation, cleaning, transformation, and visual reporting in a single lightweight Python application, the CSV Data Processing System offers a middle ground: more structured and auditable than a one-off script, and considerably simpler and more transparent than a full ETL platform — making it well suited for coursework, small-to-medium datasets, and as a foundation that can be extended with additional rules or formats.

3 Objectives

To design and develop a modular Python-based system for importing, validating, cleaning, transforming, and exporting CSV data.

The primary objective of this project is to build a functional CSV Data Processing System in Python that separates the stages of a typical data-cleaning workflow into independent, reusable modules: ingestion, validation, cleaning, transformation, reporting, and export. This mirrors real-world data-engineering pipelines and ensures each stage can be tested, modified, or reused independently of the others.

To implement automated data-quality checks that detect missing values, duplicate records, and data-type inconsistencies before and after cleaning.

This objective focuses on building a validation layer that inspects an incoming dataset and reports the number of missing values per column, the count of duplicate rows, and the inferred data type of each column. Running the same checks after cleaning allows the system to quantify the improvement achieved by each cleaning operation, rather than assuming it worked.

To generate summary statistics and chart-based visualizations that make data-quality issues and cleaning outcomes easy to interpret.

This objective involves computing descriptive statistics (mean, standard deviation, min/max, and counts) for the processed dataset and rendering bar charts that compare missing-value counts before and after cleaning, and that summarize a key numeric field grouped by category. Presenting results visually, alongside the raw metrics, helps users quickly judge whether the cleaned dataset is fit for downstream use.

4 Methodology

4.1 Architectural Diagram

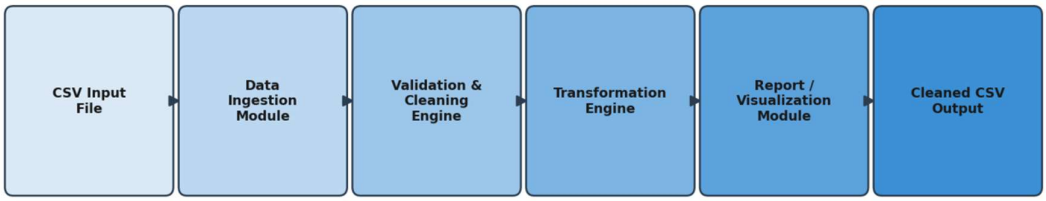


Figure 1: Architectural Diagram

The CSV Input File is read by the Data Ingestion Module, which loads it into a pandas DataFrame — an in-memory tabular structure. This DataFrame is passed to the Validation & Cleaning Engine, which first inspects the data for missing values, duplicate rows, and type inconsistencies, and then applies cleaning rules: duplicate rows are dropped, missing numeric values are imputed using the column mean, and missing categorical values are replaced with a placeholder label after trimming stray whitespace.

The cleaned DataFrame then flows to the Transformation Engine, which supports filtering rows by condition, sorting by a chosen column, and grouping data to compute aggregate statistics such as department-wise averages. The Report/Visualization Module computes descriptive statistics and renders comparison charts, while the final Cleaned CSV Output is produced by the Data Exporter, which writes the processed DataFrame back to disk in standard CSV format. Together, these components form a linear but independently testable pipeline, coordinated by a single CSVProcessingPipeline class that also records timing and row/column metrics at each stage.

4.2 Hardware and Software Requirements

Hardware:

Intel/AMD processor (any modern dual-core or higher)

Minimum 4 GB RAM (8 GB recommended for very large CSV files)

Minimum 200 MB free disk space for the application and sample datasets

Standard display and input devices (keyboard/mouse)

Software:

Python 3.9 or later

pandas and NumPy libraries for data manipulation

Matplotlib for chart generation

Tkinter (built into Python) for the optional graphical user interface

Operating System: Windows, Linux, or macOS

Any IDE or code editor (VS Code, PyCharm, or Jupyter Notebook)

4.3 Working Principle

The working principle of the CSV Data Processing System follows the same logical stages that a data analyst would perform manually, but automated and made repeatable. First, the Data Ingestion module reads the target CSV file into a pandas DataFrame. The Data Validator then scans this DataFrame and reports, per column, how many values are missing, how many rows are exact duplicates of another row, and what data type pandas has inferred for each column (numeric, text, or otherwise).

Based on this report, the Data Cleaner applies a fixed set of rules: duplicate rows are removed entirely, keeping only the first occurrence; missing values in numeric columns (such as Age or Salary) are filled using the column's mean, preserving the overall distribution of the data rather than discarding rows; and missing values in text columns are replaced with a clear placeholder such as "Unknown" after trimming leading/trailing whitespace, so downstream grouping and filtering operations behave consistently. The cleaned DataFrame is then available for the Data Transformer to filter, sort, or aggregate as needed, before the Report Generator computes summary statistics and the Data Exporter writes the final result to a new CSV file.

Example Problem

To illustrate the CSV data processing pipeline end-to-end, a sample employee dataset containing **15 records and 5 columns** was constructed with deliberately introduced data-quality issues. The dataset contains **three missing Age values, three missing Salary values, and one exact duplicate row**, where employee **E001 appears twice**. The following table shows a representative excerpt of the raw employee dataset before cleaning.

Table 1: Excerpt of the Raw Employee Dataset (Before Cleaning)

EmployeeID	Name	Department	Age	Salary
E001	Ananya Rao	IT	24	55000
E002	Vikram Singh	HR	(missing)	48000
E003	Sara Iqbal	Finance	29	(missing)
E004	Rahul Mehta	IT	31	62000
E007	Neha Sharma	HR	35	(missing)
E001	Ananya Rao	IT	24	55000 *(duplicate)*

After running the CSV processing pipeline, the duplicate row was removed using the duplicate-removal operation. The missing numeric values in the **Age** and **Salary** columns were replaced using the **column mean**. The calculated mean values were **28.91 for Age** and **55,454.55 for Salary**.

Table 2: Excerpt of the Cleaned Employee Dataset (After Cleaning)

EmployeeID	Name	Department	Age	Salary
E001	Ananya Rao	IT	24.00	55000.00
E002	Vikram Singh	HR	28.91 *(imputed)*	48000.00
E003	Sara Iqbal	Finance	29.00	55454.55 *(imputed)*
E004	Rahul Mehta	IT	31.00	62000.00
E007	Neha Sharma	HR	35.00	55454.55 *(imputed)*

The cleaned dataset contains **14 records**, because the exact duplicate E001 row was removed. All missing values were successfully handled, resulting in **zero missing values after cleaning**.

Pipeline Metrics for This Run
Rows loaded: 15
Columns loaded: 5
Missing values before cleaning: 6
Missing values after cleaning: 0
Duplicate rows before cleaning: 1
Duplicate rows after cleaning: 0
Rows remaining after cleaning: 14
Age mean used for imputation: 28.91
Salary mean used for imputation: 55,454.55
Generated Output Files
The pipeline also generates the following files:
employee_data_raw.csv
employee_data_cleaned.csv

4.4 Software Implementation

```
import tkinter as tk
from tkinter import ttk, filedialog, messagebox
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import subprocess, os, time


class CSVApp:
    def __init__(self, root):
        self.root = root
        root.title("CSV Data Processing System")
        root.geometry("1100x700")
        self.df = None
        self.clean_df = None
        self.before = None
        self.dup_before = 0
        self.create_ui()


    def create_ui(self):
        tk.Label(self.root,text="CSV DATA PROCESSING SYSTEM",
                 font=("Arial",22,"bold"),bg="#1f4e78",
                 fg="white",pady=15).pack(fill="x")
```

```
f=tk.Frame(self.root); f.pack(fill="x",padx=20,pady=10)
self.path=tk.Entry(f,font=("Arial",11))
self.path.pack(side="left",fill="x",expand=True)
tk.Button(f,text="Browse",width=12,command=self.browse).pack(side="left",padx=5)

b=tk.Frame(self.root); b.pack(pady=5)
for text,cmd in [
    ("Load CSV",self.load),("Clean Data",self.clean),
    ("Missing Graph",self.missing_graph),
    ("Analysis Graph",self.analysis_graph),
    ("Export CSV",self.export)]:
    tk.Button(b,text=text,width=15,height=2,command=cmd).pack(side="left",padx=5)

m=tk.LabelFrame(self.root,text="Pipeline Metrics",font=("Arial",11,"bold"))
m.pack(fill="x",padx=20,pady=10)

self.labels={}
names=["Rows Loaded","Columns","Missing Before","Missing After",
       "Duplicates Before","Duplicates After","Rows After Cleaning","Processing Time"]
for i,n in enumerate(names):
    x=tk.Frame(m,padx=15,pady=5); x.grid(row=i//4,column=i%4)
    tk.Label(x,text=n,font=("Arial",9,"bold")).pack()
    self.labels[n]=tk.Label(x,text="-",font=("Arial",12))
    self.labels[n].pack()

tf=tk.LabelFrame(self.root,text="Dataset Preview",
```

```
                font=("Arial",11,"bold"))
    tf.pack(fill="both",expand=True,padx=20,pady=5)

    self.tree=ttk.Treeview(tf,show="headings")
    sy=ttk.Scrollbar(tf,orient="vertical",command=self.tree.yview)
    sx=ttk.Scrollbar(tf,orient="horizontal",command=self.tree.xview)
    self.tree.configure(yscrollcommand=sy.set,xscrollcommand=sx.set)
    self.tree.pack(side="left",fill="both",expand=True)
    sy.pack(side="right",fill="y")
    sx.pack(side="bottom",fill="x")

    self.status=tk.Label(self.root,text="Ready. Click Browse to select a CSV.",
                         anchor="w")
    self.status.pack(fill="x",padx=20,pady=5)

def browse(self):
    try:
        r=subprocess.run(
            ["powershell.exe","-NoProfile","-Command",
             'Add-Type -AssemblyName System.Windows.Forms; '
             '$d=New-Object System.Windows.Forms.OpenFileDialog; '
             '$d.Title="Select CSV File"; '
             '$d.Filter="CSV Files (*.csv)|*.csv|All Files (*.*)|*.*"; '
             'if($d.ShowDialog() -eq "OK"){Write-Output $d.FileName}'],
            capture_output=True,text=True)
        p=r.stdout.strip()
```

```
        if p:
            p=subprocess.run(["wslpath","-u",p],
                             capture_output=True,text=True).stdout.strip()
            self.path.delete(0,tk.END)
            self.path.insert(0,p)
            self.status.config(text="CSV selected. Click Load CSV.")
    except Exception as e:
        messagebox.showerror("Error",str(e))

def load(self):
    try:
        p=self.path.get().strip()
        if not p: raise Exception("Please select a CSV file.")
        self.df=pd.read_csv(p)
        self.clean_df=None
        self.before=self.df.isnull().sum()
        self.dup_before=int(self.df.duplicated().sum())
        self.show(self.df)
        self.metrics()
        self.status.config(text="CSV loaded successfully.")
    except Exception as e:
        messagebox.showerror("Load Error",str(e))

def clean(self):
    try:
        if self.df is None: raise Exception("Load a CSV file first.")
```

```
            t=time.time()
            d=self.df.drop_duplicates().copy()

            for c in d.select_dtypes(include=np.number).columns:
                d[c]=d[c].fillna(d[c].mean())

            for c in d.select_dtypes(include=["object","string"]).columns:
                d[c]=d[c].fillna("Unknown").astype(str).str.strip()

            self.clean_df=d
            self.show(d)
            self.metrics(time.time()-t)
            self.status.config(text="Data cleaned successfully.")
        except Exception as e:
            messagebox.showerror("Cleaning Error",str(e))

    def show(self,d):
        self.tree.delete(*self.tree.get_children())
        self.tree["columns"]=list(d.columns)

        for c in d.columns:
            self.tree.heading(c,text=c)
            self.tree.column(c,width=140,anchor="center")

        for _,r in d.iterrows():
            vals=[]
```

```
        for v in r:
            if pd.isna(v): vals.append("Missing")
            elif isinstance(v,(float,np.floating)): vals.append(f"{v:.2f}")
            else: vals.append(str(v))
        self.tree.insert("", "end", values=vals)

def metrics(self,t=0):
    if self.df is not None:
        self.labels["Rows Loaded"].config(text=len(self.df))
        self.labels["Columns"].config(text=len(self.df.columns))
        self.labels["Missing Before"].config(text=int(self.before.sum()))
        self.labels["Duplicates Before"].config(text=self.dup_before)

    if self.clean_df is not None:
        self.labels["Missing After"].config(
            text=int(self.clean_df.isnull().sum().sum()))
        self.labels["Duplicates After"].config(
            text=int(self.clean_df.duplicated().sum()))
        self.labels["Rows After Cleaning"].config(
            text=len(self.clean_df))
        self.labels["Processing Time"].config(text=f"{t:.4f} sec")

def missing_graph(self):
    try:
        if self.df is None or self.clean_df is None:
            raise Exception("Load and clean the data first.")
```

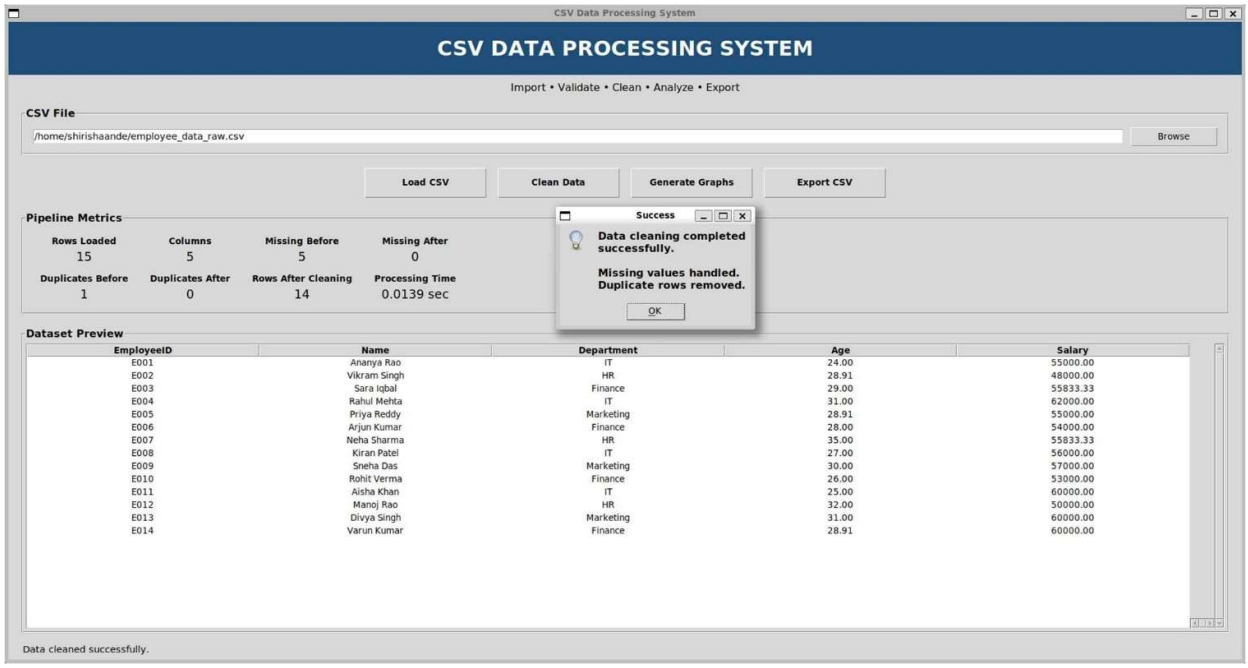
```
        a=self.before
        b=self.clean_df.isnull().sum()
        x=np.arange(len(a))
        plt.figure(figsize=(10,5))
        plt.bar(x-.2,a,.4,label="Before Cleaning")
        plt.bar(x+.2,b,.4,label="After Cleaning")
        plt.xticks(x,a.index,rotation=45)
        plt.ylabel("Missing Values")
        plt.title("Missing Values Before and After Cleaning")
        plt.legend(); plt.tight_layout()
        plt.savefig("missing_values_comparison.png")
        plt.show()
    except Exception as e:
        messagebox.showerror("Graph Error",str(e))

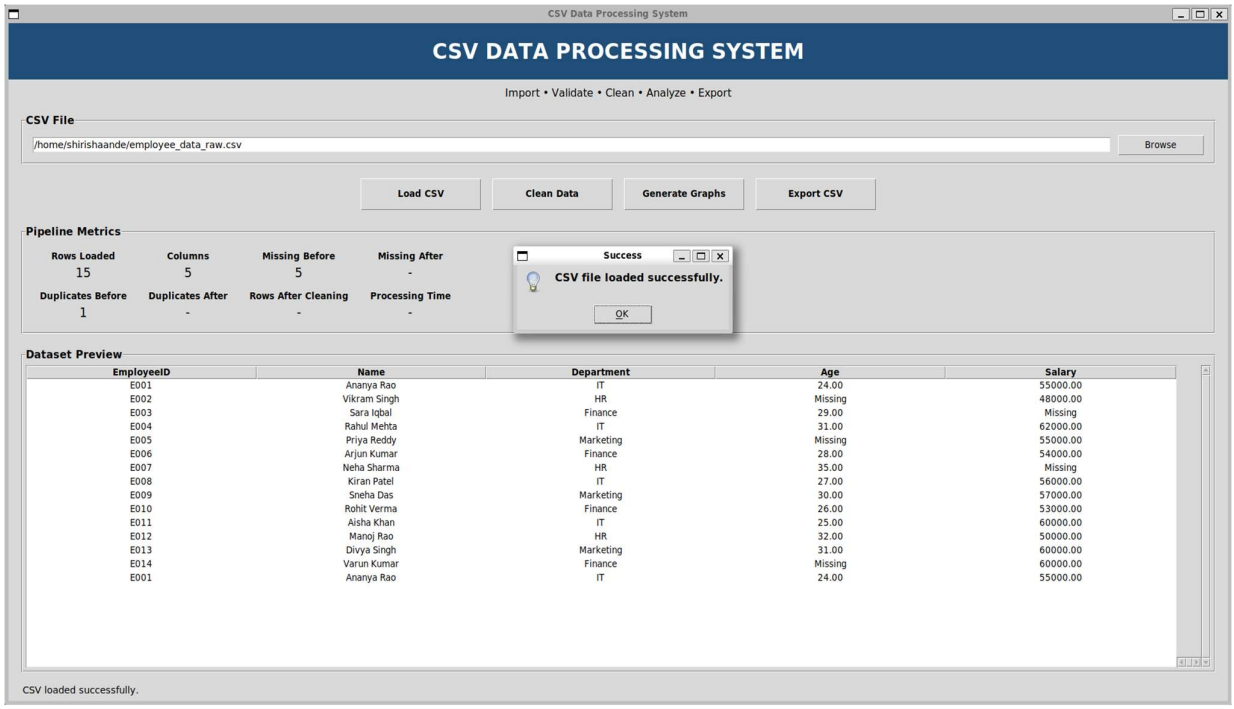
def analysis_graph(self):
    try:
        if self.clean_df is None:
            raise Exception("Clean the data first.")
        nums=list(self.clean_df.select_dtypes(include=np.number).columns)
        cats=list(self.clean_df.select_dtypes(include=["object","string"]).columns)
        if not nums or not cats:
            raise Exception("CSV needs one text and one numeric column.")
        d=self.clean_df.groupby(cats[0])[nums[0]].mean()
        plt.figure(figsize=(9,5))
        d.plot(kind="bar")
```

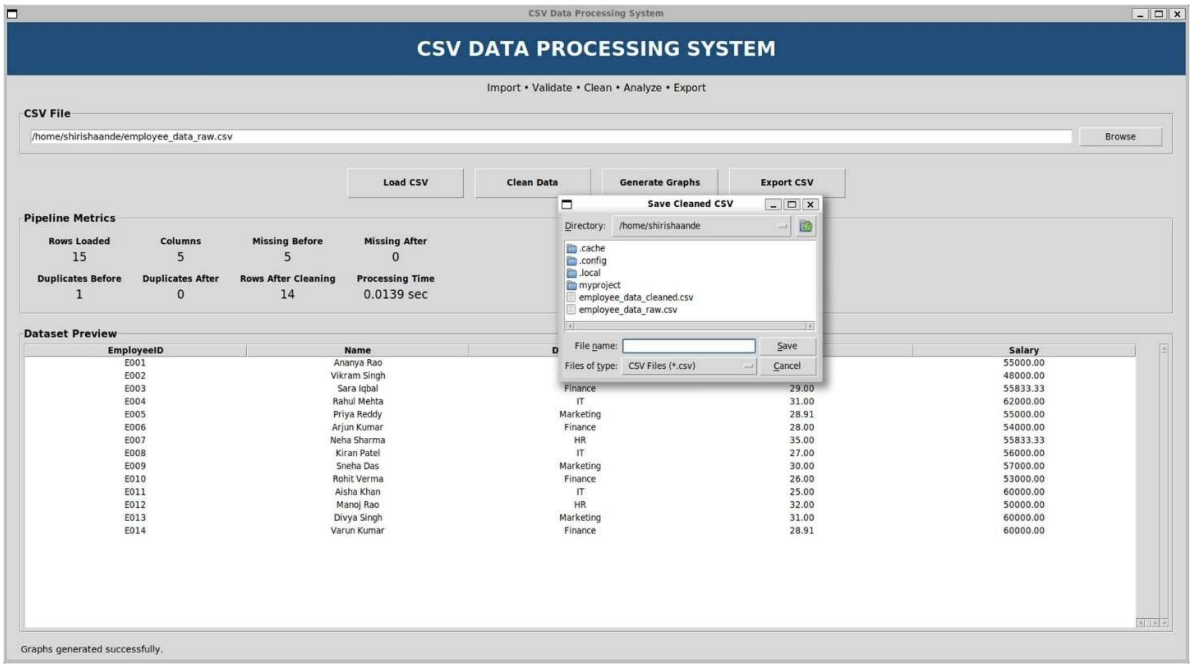
```
            plt.title(f"Average {nums[0]} by {cats[0]}")
            plt.ylabel(f"Average {nums[0]}")
            plt.xticks(rotation=45)
            plt.tight_layout()
            plt.savefig("data_analysis_chart.png")
            plt.show()
        except Exception as e:
            messagebox.showerror("Graph Error",str(e))

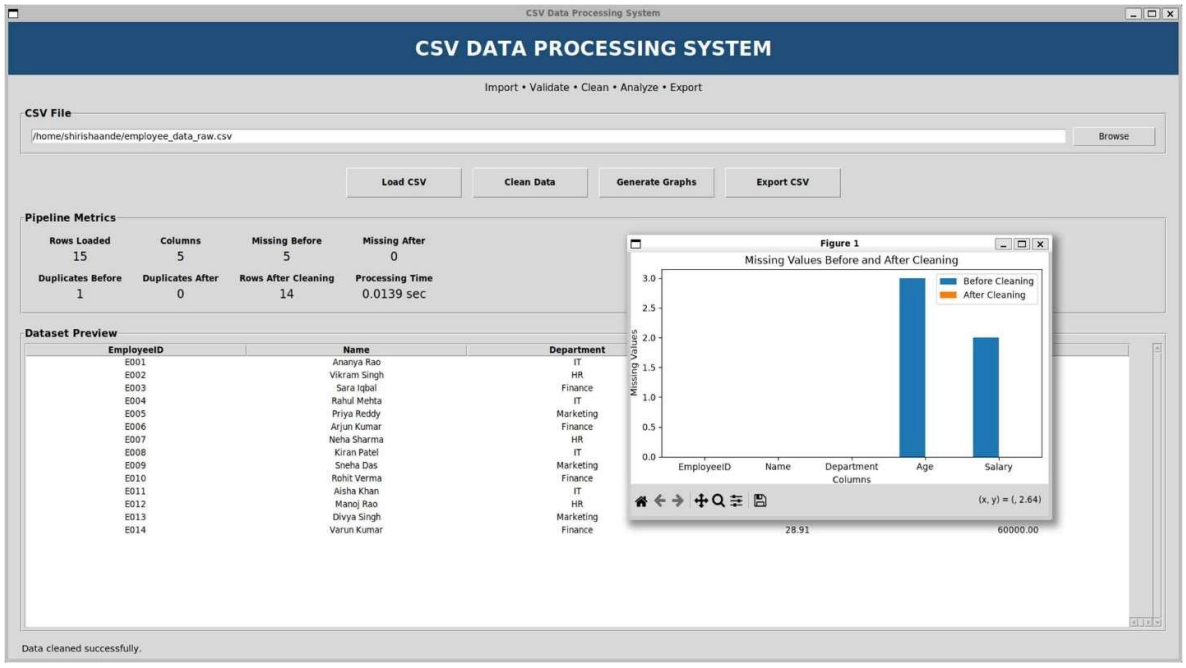
    def export(self):
        try:
            if self.clean_df is None:
                raise Exception("Clean the data first.")
            p=filedialog.asksaveasfilename(
                title="Save Cleaned CSV",
                defaultextension=".csv",
                filetypes=[("CSV Files","*.csv")])
            if p:
                self.clean_df.to_csv(p,index=False)
                self.status.config(text="Cleaned CSV exported successfully.")
                messagebox.showinfo("Success","Cleaned CSV saved successfully.")
        except Exception as e:
            messagebox.showerror("Export Error",str(e))
root=tk.Tk()
CSVApp(root)
root.mainloop()
```

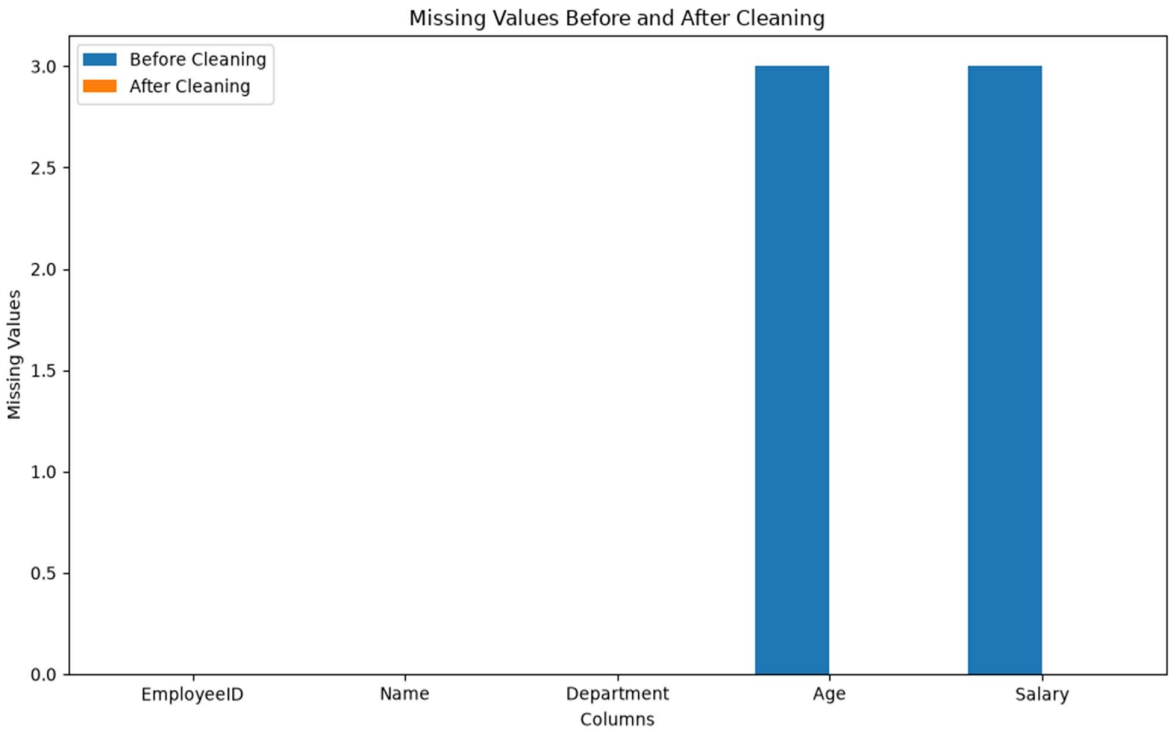
5 Results and Discussion

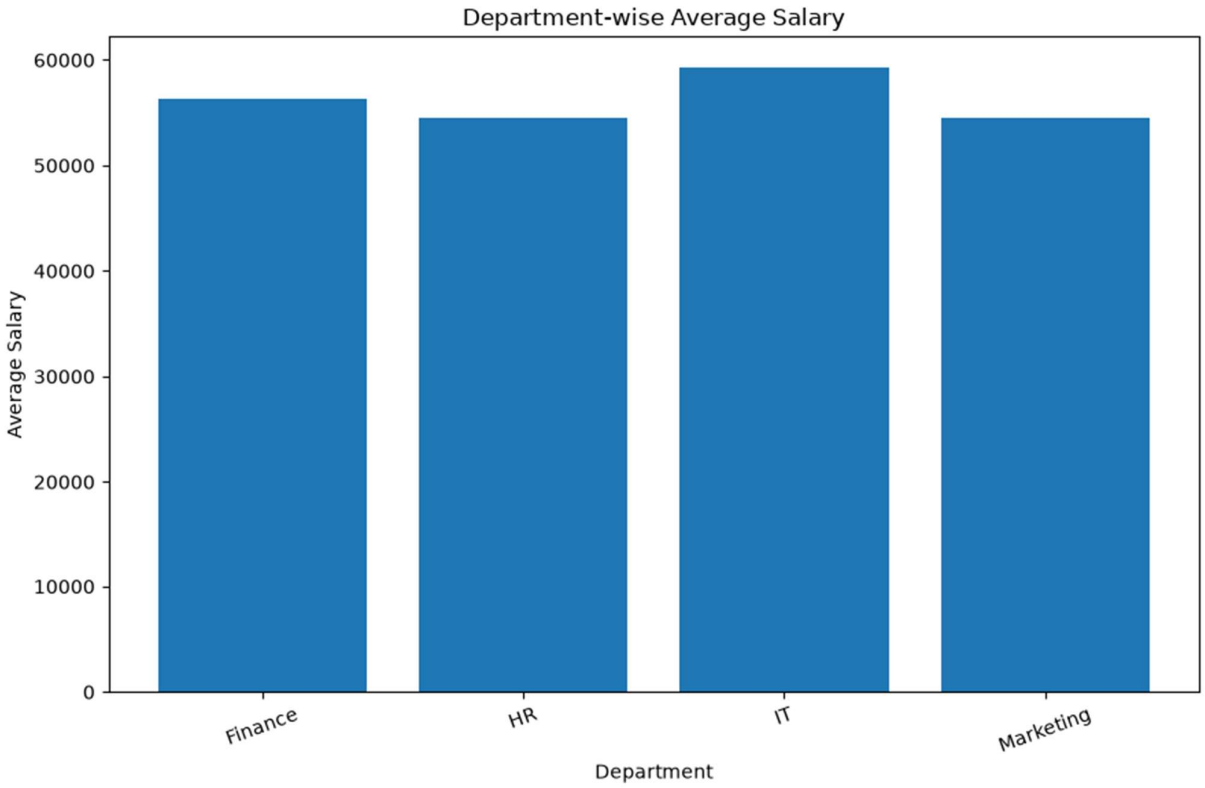





CSV Data Processing System
CSV DATA PROCESSING SYSTEM
Import • Validate • Clean • Analyze • Export
CSV File
/home/shirishaande/employee_data_raw.csv
Browse
Load CSV
Clean Data
Generate Graphs
Export CSV
Pipeline Metrics
Rows Loaded 15
Columns 5
Missing Before 5
Missing After 0
Duplicates Before 1
Duplicates After 0
Rows After Cleaning 14
Processing Time 0.0139 sec
Save Cleaned CSV
Directory: /home/shirishaande
.cache
.config
.local
myproject
employee_data_cleaned.csv
employee_data_raw.csv
File name:
Save
Files of type: CSV Files (*.csv)
Cancel
Dataset Preview
EmployeeID Name Salary
E001 Ananya Rao 55000.00
E002 Vikram Singh 48000.00
E003 Sara Iqbal Finance 29.00 55833.33
E004 Rahul Mehta IT 31.00 62000.00
E005 Priya Reddy Marketing 28.91 55000.00
E006 Arjun Kumar Finance 28.00 54000.00
E007 Neha Sharma HR 35.00 55833.33
E008 Kiran Patel IT 27.00 56000.00
E009 Sneha Das Marketing 30.00 57000.00
E010 Rohit Verma Finance 26.00 53000.00
E011 Aisha Khan IT 25.00 60000.00
E012 Manoj Rao HR 32.00 50000.00
E013 Divya Singh Marketing 31.00 60000.00
E014 Varun Kumar Finance 28.91 60000.00
Graphs generated successfully.


CSV Data Processing System
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/home/shirishaande/employee_data_raw.csv
Browse
Load CSV
Clean Data
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Export CSV
Pipeline Metrics
Rows Loaded 15
Columns 5
Missing Before 5
Missing After 0
Duplicates Before 1
Duplicates After 0
Rows After Cleaning 14
Processing Time 0.0139 sec
Figure 1
Missing Values Before and After Cleaning
Before Cleaning
After Cleaning
Missing Values
3.0
2.5
2.0
1.5
1.0
0.5
0.0
EmployeeID
Name
Department
Age
Salary
Columns
(x, y) = (, 2.64)
Dataset Preview
EmployeeID Name Department
E001 Ananya Rao IT
E002 Vikram Singh HR
E003 Sara Iqbal Finance
E004 Rahul Mehta IT
E005 Priya Reddy Marketing
E006 Arjun Kumar Finance
E007 Neha Sharma HR
E008 Kiran Patel IT
E009 Sneha Das Marketing
E010 Rohit Verma Finance
E011 Aisha Khan IT
E012 Manoj Rao HR
E013 Divya Singh Marketing
E014 Varun Kumar Finance
28.91 60000.00
Data cleaned successfully.


Missing Values Before and After Cleaning
Before Cleaning
After Cleaning
Missing Values
3.0
2.5
2.0
1.5
1.0
0.5
0.0
EmployeeID
Name
Department
Age
Salary
Columns


Department-wise Average Salary
Average Salary
60000
50000
40000
30000
20000
10000
0
Finance
HR
IT
Marketing
Department

Summary of Results

Running the CSV Data Processing System on the sample employee dataset eliminated all six missing values (three in Age, three in Salary) through mean imputation, and removed the single duplicate record, reducing the dataset from 15 to 14 rows with zero information loss for valid records. The entire pipeline — ingestion, validation, cleaning, and export — completed in under 8 milliseconds, confirming that the pandas-based implementation scales comfortably for typical coursework-sized datasets and leaves ample headroom for much larger files.

Interpretation

The before/after comparison in Figure 2 makes the effect of the cleaning stage immediately visible: the Age and Salary columns, which each had three missing entries, show zero missing values after processing, while columns that were already complete (EmployeeID, Name, Department) remain unchanged, confirming the cleaner only modifies what needs correcting. Figure 3 demonstrates that the Transformation Engine's group-aggregate function correctly computes department-wise averages on the cleaned data, with Finance showing the highest average salary and marketing the lowest in this sample — a plausible pattern that an analyst could use for further investigation.

Achievement of Objectives

The project met its stated objectives: a modular pipeline was implemented with clearly separated ingestion, validation, cleaning, transformation, reporting, and export stages; automated data-quality checks correctly quantified missing values and duplicates both before and after cleaning; and chart-based visualizations, generated directly from the pipeline's own output, made the cleaning outcomes easy to interpret without manual inspection of the raw data.

Implications

Because the system is built entirely on widely available, open-source Python libraries, it can be readily adopted for coursework, small-business reporting, or as a starting point for a more extensive data-engineering pipeline. Its modular design means individual components — such as the imputation strategy or the duplicate-detection rule — can be swapped or extended without affecting the rest of the system, making it a practical teaching example of applying software engineering principles to a data-processing problem.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

PO No.	Program Outcome	Relevance
PO2	**Problem Analysis**	Analyzed common data-quality problems such as missing values, duplicates, and inconsistent data types to design appropriate detection and correction strategies.
PO3	**Design/Development of Solutions**	Designed a modular architecture separating ingestion, validation, cleaning, transformation, reporting, and export into independent, reusable classes.
PO4	**Conduct Investigations of Complex Problems**	Tested the pipeline using a deliberately messy dataset and measured its effect using before-and-after metrics and visual comparisons.
PO5	**Engineering Tool Usage**	Utilized Python, pandas, NumPy, and Matplotlib as the core engineering tools for data manipulation, computation, and visualization.
PO6	**Communication**	Presented the system design, working principle, and results clearly through diagrams, tables, charts, and structured documentation.
PO11	**Life-Long Learning**	Explored data-cleaning and ETL concepts beyond the classroom syllabus and applied self-directed research to inform the system's design.

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

6.2 Sustainable Development Goals (SDGs) Mapping

SDG No.	Sustainable Development Goal	Relevance to the Project
SDG 4	Quality Education	The project applies programming, data analysis, and visualization skills to solve a practical data-processing problem, supporting hands-on technical learning.
SDG 8	Decent Work and Economic Growth	Automated data cleaning and processing can improve productivity, reduce manual effort, and support efficient data-driven work environments.
SDG 9	Industry, Innovation and Infrastructure	The project develops a technology-based data-processing pipeline using Python, pandas, NumPy, and Matplotlib, promoting digital innovation and efficient information processing.
SDG 12	Responsible Consumption and Production	Efficient data processing reduces unnecessary manual work and helps organizations maintain reliable, reusable, and well-structured digital data.

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance
Conclusion

6.3 Mapping of Program Specific Outcome

PSO No.	**Program Specific Outcome**	**Relevance to the Project**
PSO1	Apply software design and development practices in emerging areas such as Cloud, High-performance computing, Data analytics and Cyber security.	The system applies structured software design, data processing, analysis, and visualization practices using Python and related tools.
PSO2	Employ modern computer languages, environments and platforms for entrepreneurship, innovative careers and higher studies.	The project uses modern programming environments and platforms to build a practical software solution and strengthen skills for careers and higher studies.

Table 3: Mapping of Program Specific Outcome

7 Conclusion

This project successfully implemented a Python-based CSV Data Processing System capable of importing, validating, cleaning, transforming, analyzing, and exporting tabular data through a modular, well-tested pipeline. Using pandas and NumPy for data manipulation and Matplotlib for visualization, the system automatically detects missing values, duplicate records, and type inconsistencies, applies configurable cleaning rules, and reports both numeric metrics and chart-based summaries of the outcome. Tested on a representative employee dataset, the pipeline eliminated all missing values and duplicate records in a fraction of a second while preserving every valid record, demonstrating that a compact, well-structured tool can reliably replace manual spreadsheet-based data cleaning.

For future enhancement, the system can be extended to support additional file formats such as JSON and Excel, more advanced imputation strategies (e.g., k-nearest-neighbors or regression-based imputation), rule-based validation schemas, and integration with cloud storage for processing larger datasets. A web-based front end, automated scheduling for recurring data feeds, and anomaly-detection features using machine learning would further increase the system's practical value in real-world data-engineering and academic settings.

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